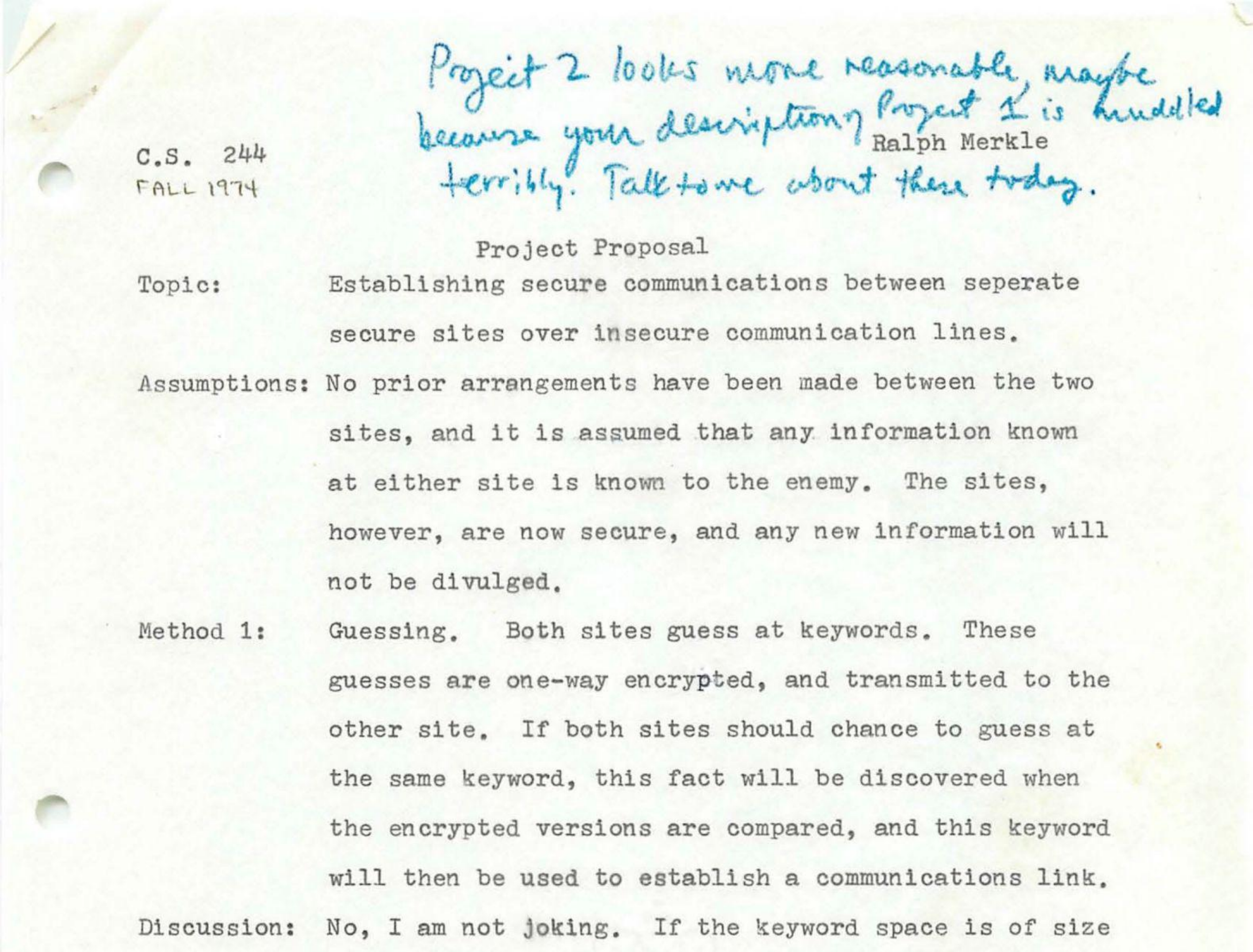
Public-Key Encryption

CS 161 Spring 2026 - Lecture 13

Announcements

- Course feedback!
  - Submit any time: henrycg.com/feedback
- Zoom is broken
- Will try to remember repeat audience questions



Project 2 looks more reasonable, maybe because your description of Project 1 is muddled terribly. Talk to me about these today.

C.S. 244
FALL 1974

Ralph Merkle

Project Proposal

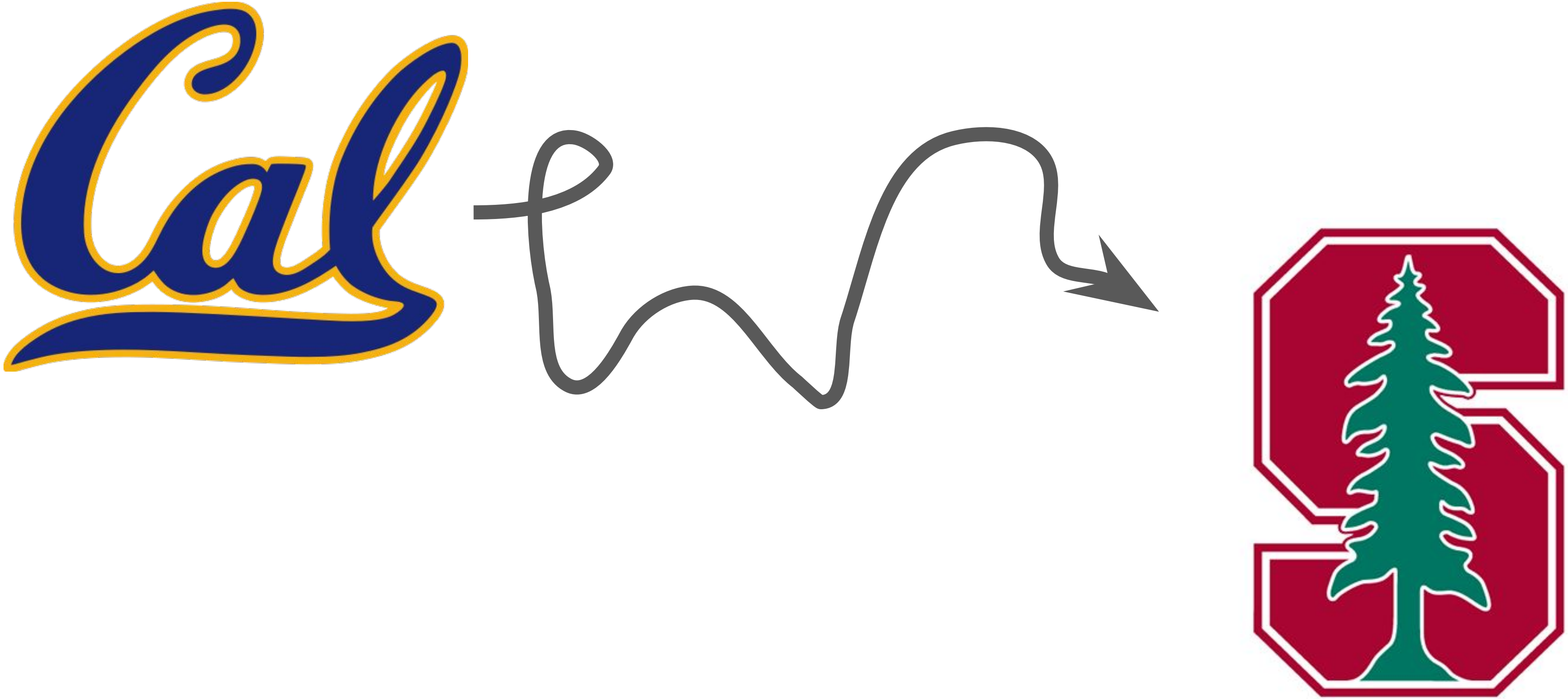
Topic: Establishing secure communications between seperate secure sites over insecure communication lines.

Assumptions: No prior arrangements have been made between the two sites, and it is assumed that any information known at either site is known to the enemy. The sites, however, are now secure, and any new information will not be divulged.

Method 1: Guessing. Both sites guess at keywords. These guesses are one-way encrypted, and transmitted to the other site. If both sites should chance to guess at the same keyword, this fact will be discovered when the encrypted versions are compared, and this keyword will then be used to establish a communications link.

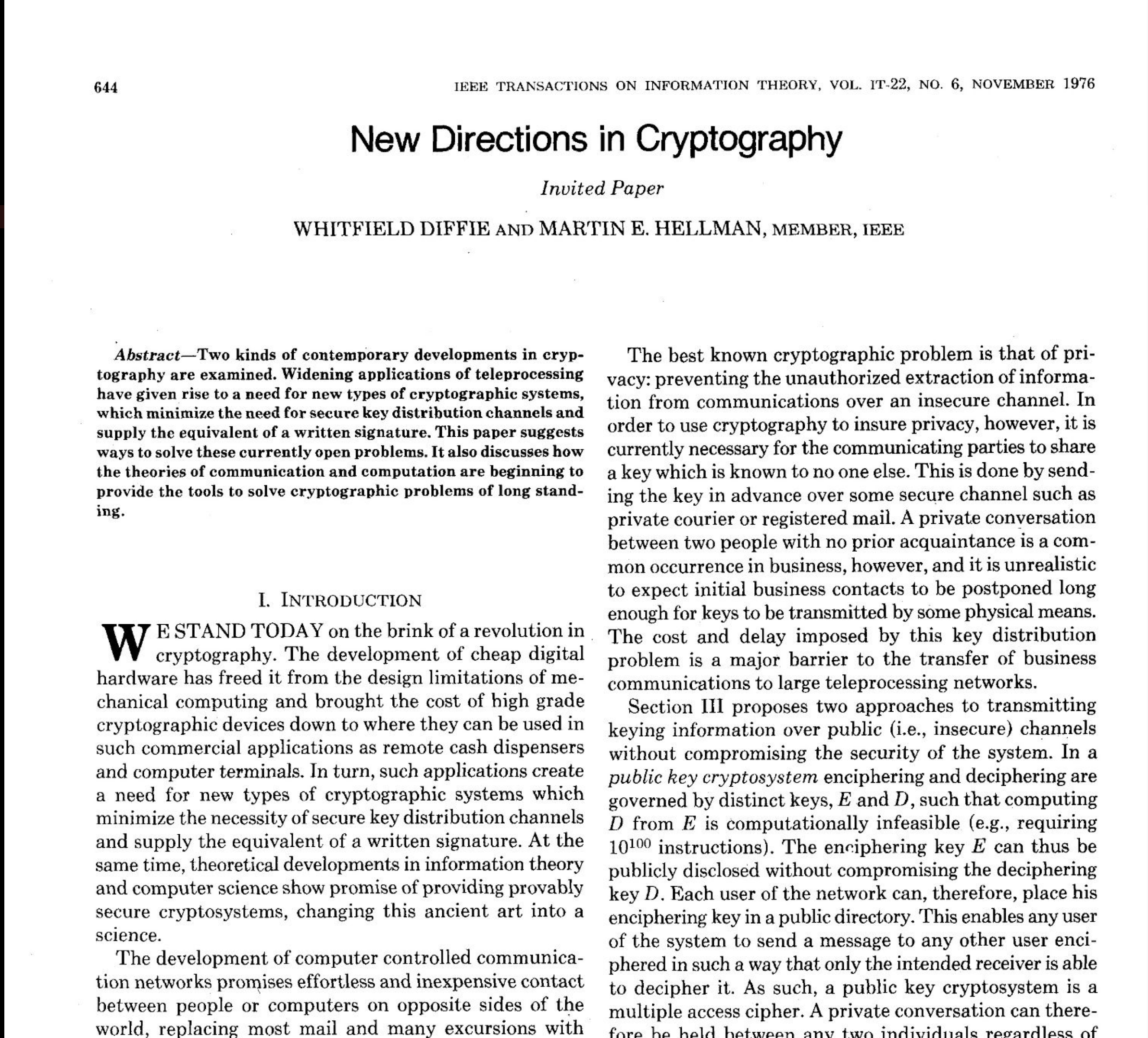
Discussion: No, I am not joking. If the keyword space is of size

And then…



Recap: Diffie-Hellman Key Exchange

- Key exchange: Agree on a shared secret over a public channel
  - Merkle puzzles: Alice/Bob = **time $O(N)$** Attacker: **time N^2**
  - Diffie-Hellman Alice/Bob = **time poly(N)** Attacker: **time $2^{N^{\wedge}\{⅓\}}$ or $2^{N/2}$**
- Diffie-Hellman algorithm:
  - Alice chooses a. Sends g^a mod p to Bob
  - Bob chooses b. Sends g^b mod p to Alice
  - Shared secret is $(g^a)^b = (g^b)^a = g^{ab}$ mod p
- Unauth. key exchange does not protect against against active attacks



644 IEEE TRANSACTIONS ON INFORMATION THEORY, VOL. IT-22, NO. 6, NOVEMBER 1976

New Directions in Cryptography

Invited Paper

WHITFIELD DIFFIE AND MARTIN E. HELLMAN, MEMBER, IEEE

***Abstract*—Two kinds of contemporary developments in cryptography are examined. Widening applications of teleprocessing have given rise to a need for new types of cryptographic systems, which minimize the need for secure key distribution channels and supply the equivalent of a written signature. This paper suggests ways to solve these currently open problems. It also discusses how the theories of communication and computation are beginning to provide the tools to solve cryptographic problems of long standing.**

I. INTRODUCTION

WE STAND TODAY on the brink of a revolution in cryptography. The development of cheap digital hardware has freed it from the design limitations of mechanical computing and brought the cost of high grade cryptographic devices down to where they can be used in such commercial applications as remote cash dispensers and computer terminals. In turn, such applications create a need for new types of cryptographic systems which minimize the necessity of secure key distribution channels and supply the equivalent of a written signature. At the same time, theoretical developments in information theory and computer science show promise of providing provably secure cryptosystems, changing this ancient art into a science.

The development of computer controlled communication networks promises effortless and inexpensive contact between people or computers on opposite sides of the world, replacing most mail and many excursions with

The best known cryptographic problem is that of privacy: preventing the unauthorized extraction of information from communications over an insecure channel. In order to use cryptography to insure privacy, however, it is currently necessary for the communicating parties to share a key which is known to no one else. This is done by sending the key in advance over some secure channel such as private courier or registered mail. A private conversation between two people with no prior acquaintance is a common occurrence in business, however, and it is unrealistic to expect initial business contacts to be postponed long enough for keys to be transmitted by some physical means. The cost and delay imposed by this key distribution problem is a major barrier to the transfer of business communications to large teleprocessing networks.

Section III proposes two approaches to transmitting keying information over public (i.e., insecure) channels without compromising the security of the system. In a *public key cryptosystem* enciphering and deciphering are governed by distinct keys, E and D, such that computing D from E is computationally infeasible (e.g., requiring 10^{100} instructions). The enciphering key E can thus be publicly disclosed without compromising the deciphering key D. Each user of the network can, therefore, place his enciphering key in a public directory. This enables any user of the system to send a message to any other user enciphered in such a way that only the intended receiver is able to decipher it. As such, a public key cryptosystem is a multiple access cipher. A private conversation can therefore be held between any two individuals regardless of

“New Directions in Cryptography” - Diffie & Hellman

The foundational paper in modern cryptography

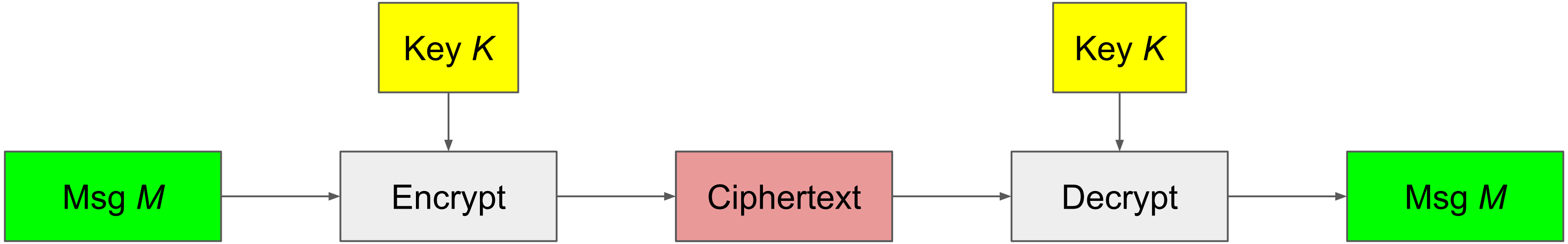
- Gave the first “secure” key-exchange scheme
- Defined public-key encryption
- Defined digital signatures

99.99% of the public-key cryptography used in practice

⇒ You must read the first page

First construction of public-key encryption and signatures was later:
– RSA = Rivest, Shamir, Adleman (1977)

WHO SAID YOU HAVE TO ENCRYPT AND DECRYPT WITH THE SAME KEY????



Key *K*

Key *K*

Msg *M*

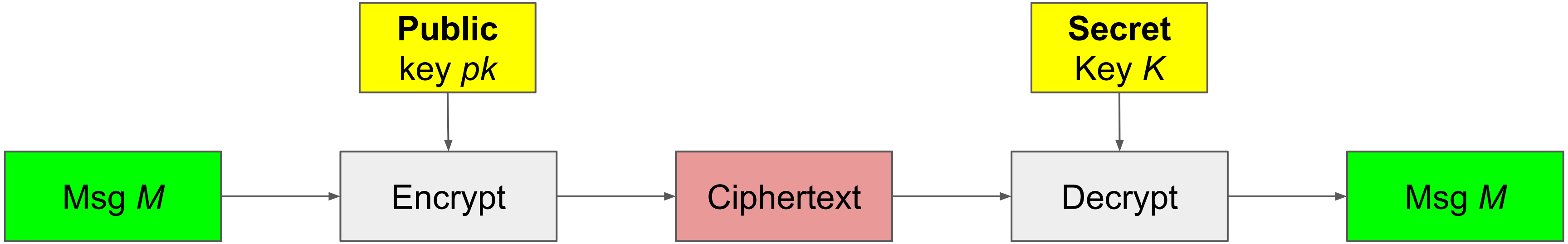
Encrypt

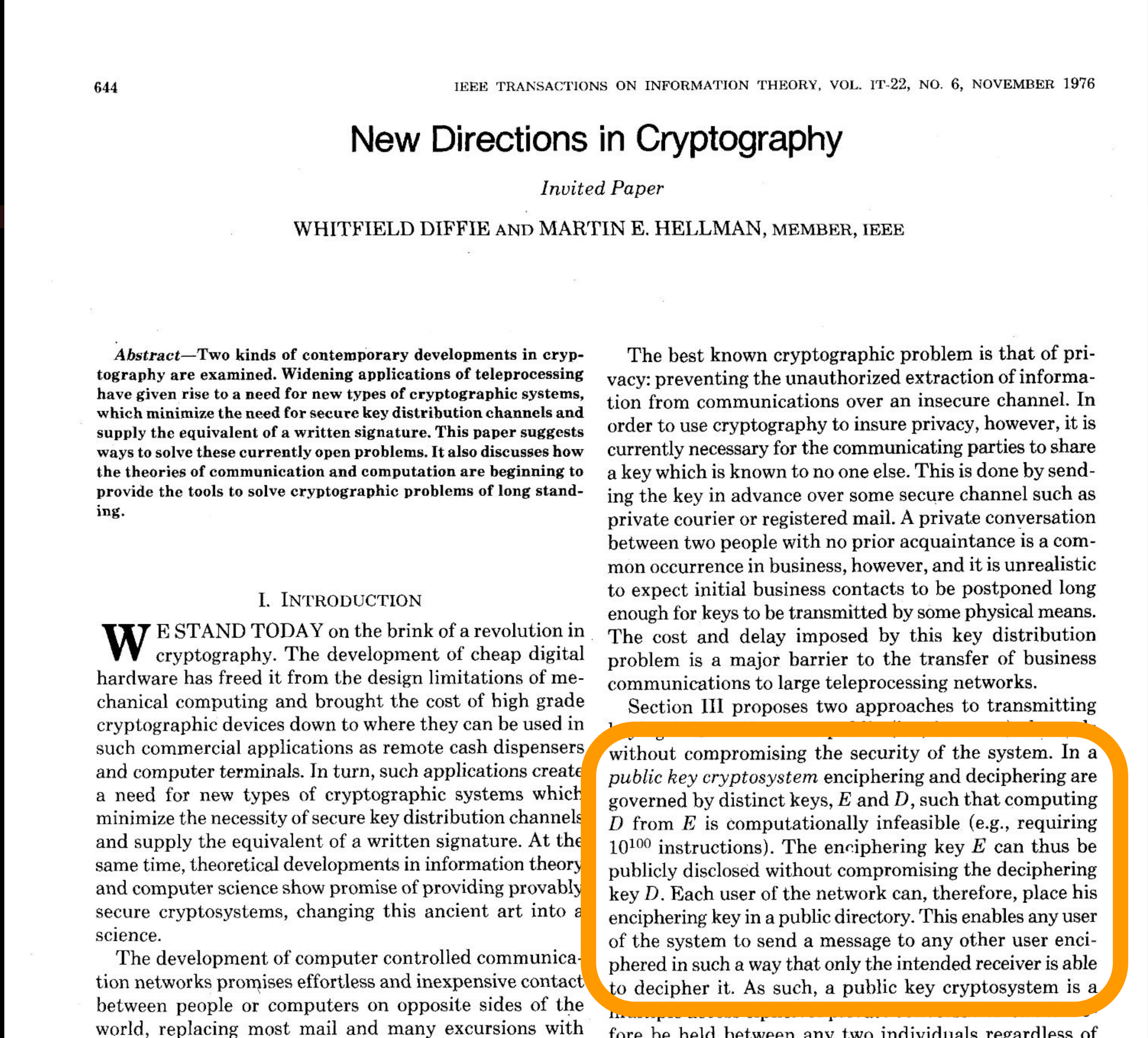
Ciphertext

Decrypt

Msg *M*

New way


Public key *pk*
Secret Key *K*
Msg *M*
Encrypt
Ciphertext
Decrypt
Msg *M*



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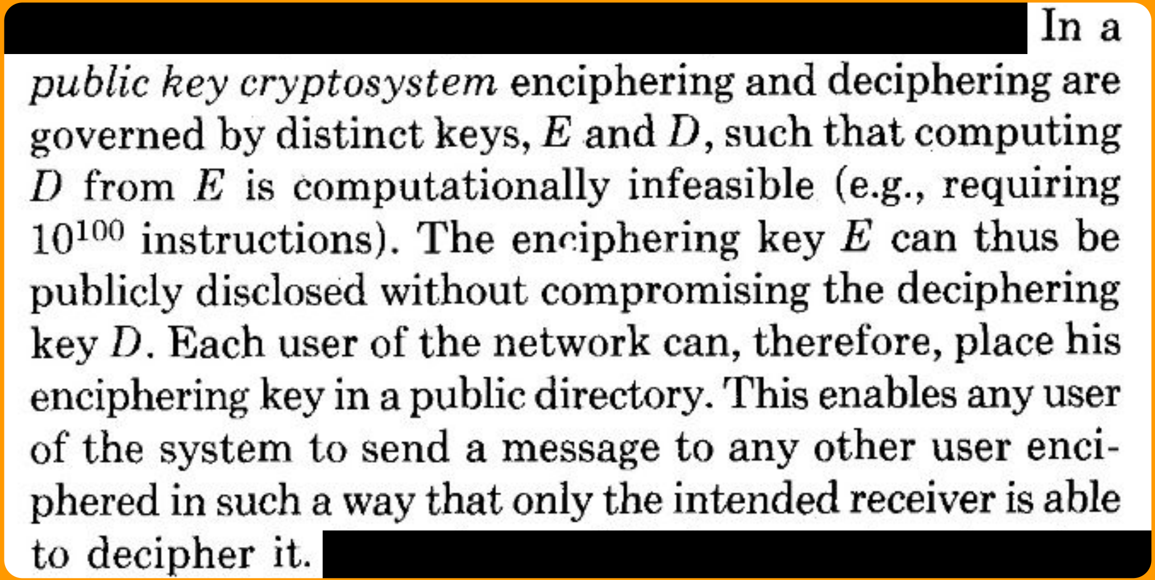

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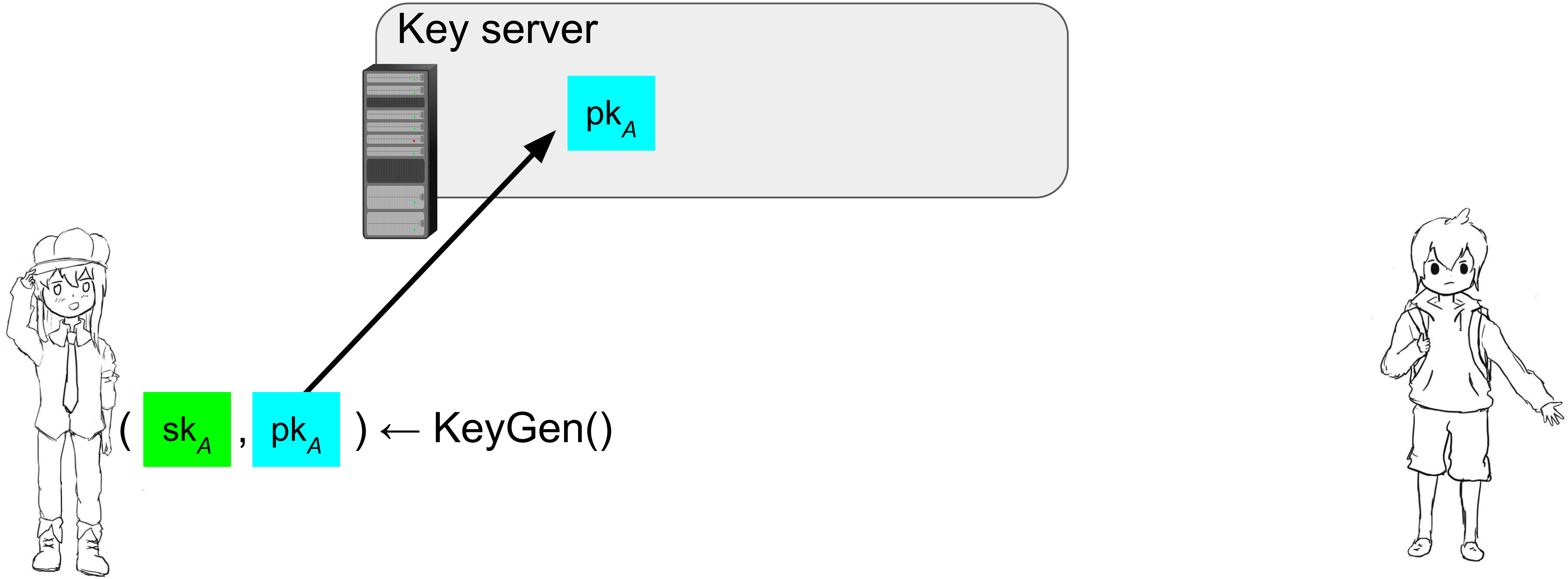
In retrosp ecret

LOCKED OUT OF CAR?
WE OPEN LOCKED DOORS! I MAKE NEW CAR KEYS!
MOBILE LOCKSMITH
DISCOUNT
LOCKSMITH

Public-Key Encryption

Textbook Chapter 11

The vision of public-key encryption (Diffie and Hellman)

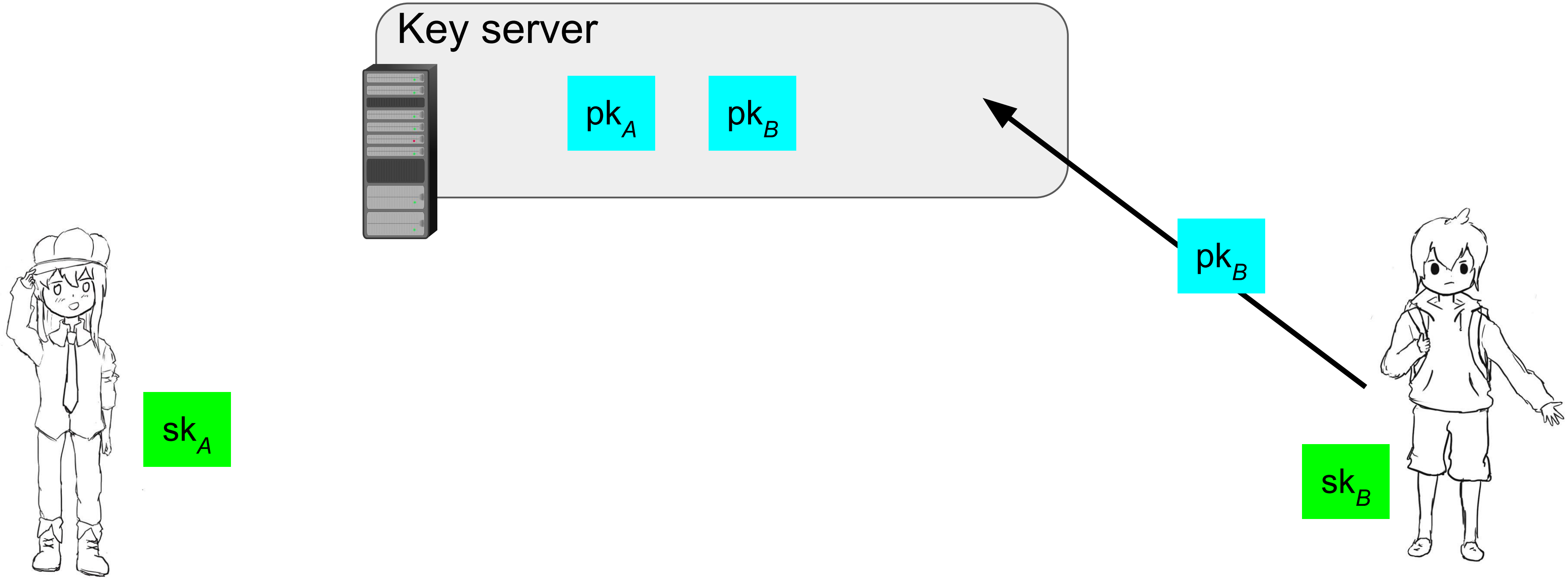


Key server

pk_A

$(sk_A, pk_A) \leftarrow \text{KeyGen}()$

The vision of public-key encryption



Key server

pk_A

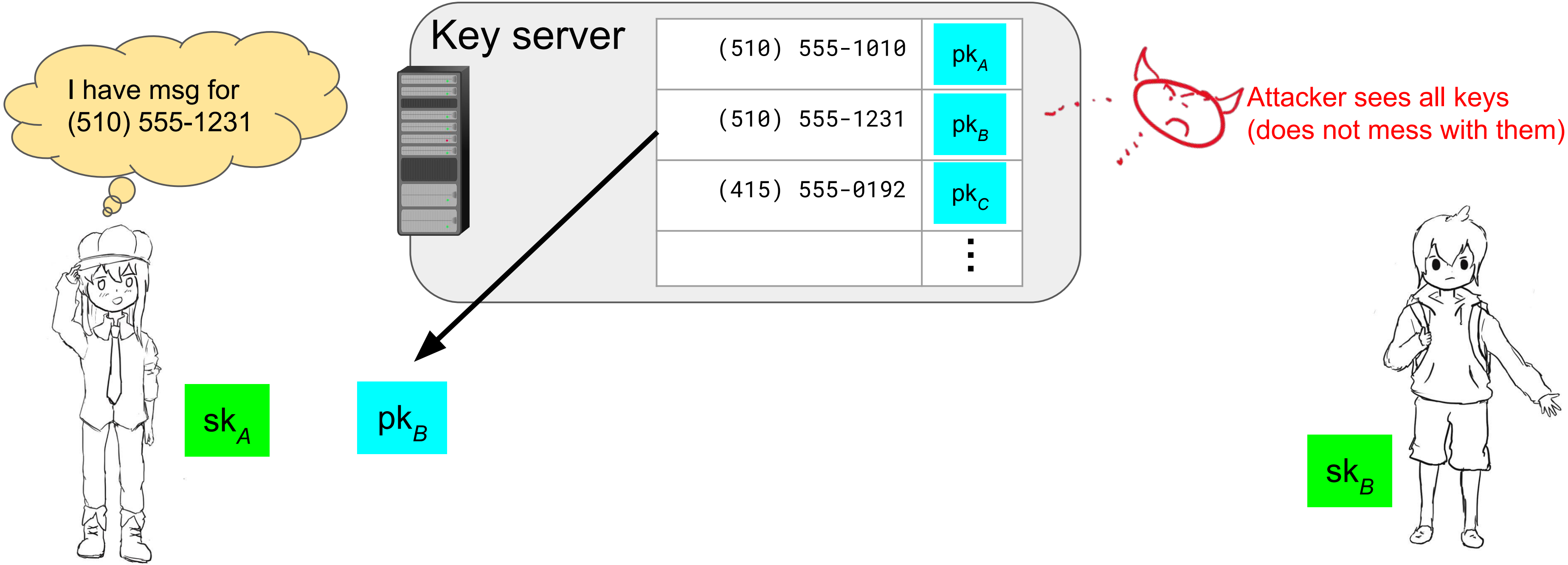
pk_B

pk_B

sk_A

sk_B

The vision of public-key encryption



I have msg for (510) 555-1231

Key server

(510) 555-1010	pk_A
(510) 555-1231	pk_B
(415) 555-0192	pk_C
	⋮

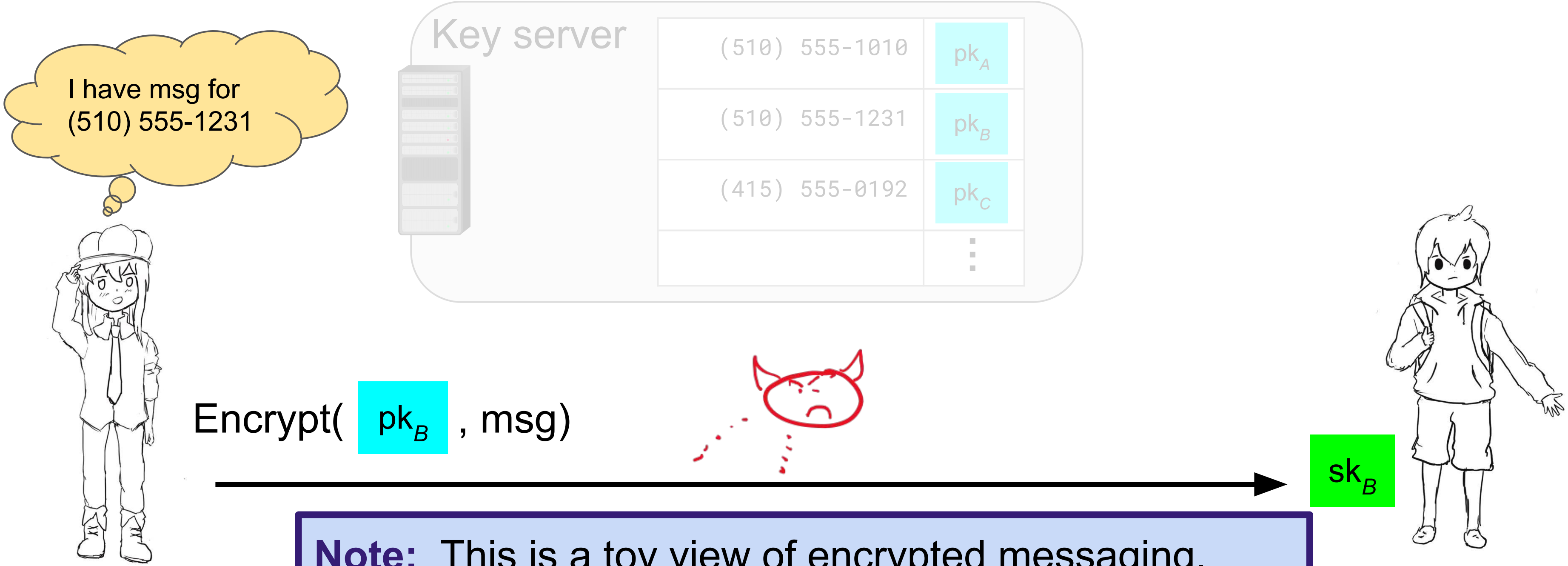
Attacker sees all keys
(does not mess with them)

sk_A

pk_B

sk_B

The vision of public-key encryption



I have msg for
(510) 555-1231

Key server

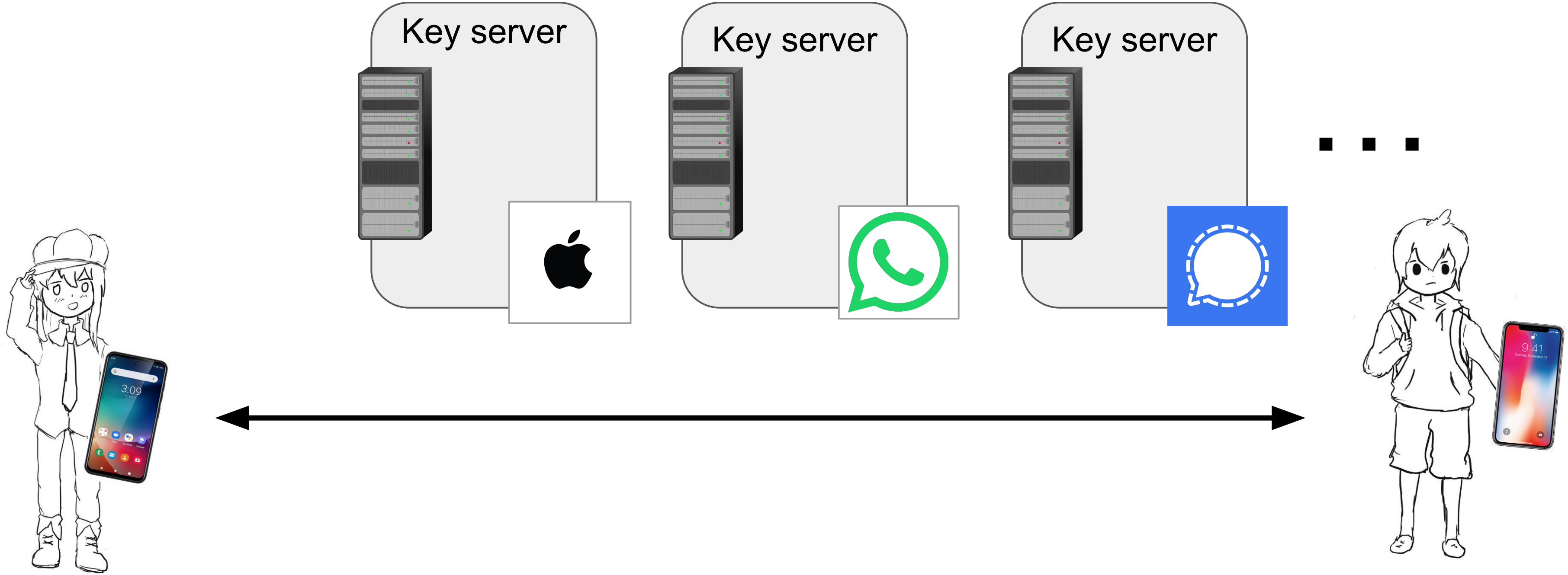
(510) 555-1010	pk_A
(510) 555-1231	pk_B
(415) 555-0192	pk_C
	⋮

Encrypt(pk_B , msg)

sk_B

Note: This is a toy view of encrypted messaging.
Real systems are much more complicated.

Public-key encryption today



Key server

Key server

Key server

. . .

Public-Key Encryption: Definition

Three efficient algorithms:

- KeyGen() → (*SK*, *PK*) // Generate keypair
- Enc(*PK*, *M*) → *C* // Encrypt with public key
- Dec(*SK*, *C*) → *M* // Decrypt with secret key

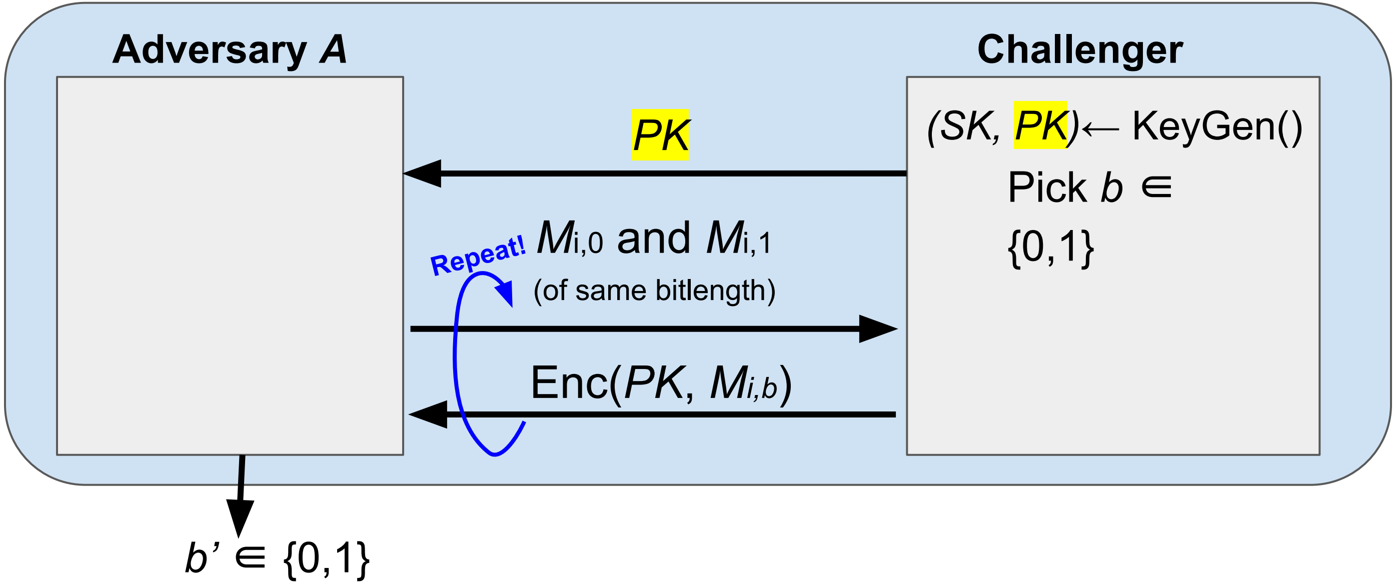
Properties:

- **Correctness**: Encrypting and then decrypting gives back the same message
  - For all (*SK*, *PK*) ← KeyGen() and messages *M*, we have: Dec(*SK*, Enc(*PK*, *M*)) = *M*
- **CPA security**:

  – Attacker gets public key and encryptions of many messages of its choice

  – Still for all equal-length M_0 and M_1 , Enc(pk, M_0) ≈ Enc(pk, M_1)

Security against chosen plaintext attack (“CPA security”)



Adversary *A*

Challenger

PK

$(SK, PK) \leftarrow \text{KeyGen}()$

Pick $b \in \{0,1\}$

Repeat!

$M_{i,0}$ and $M_{i,1}$

(of same bitlength)

$\text{Enc}(PK, M_{i,b})$

$b' \in \{0,1\}$

A public-key encryption scheme is *CPA secure* if,
for all “efficient” adversaries *A*, $|\Pr[A \text{ outputs } b] - ½| \leq$ “negligible.”

ElGamal Encryption

Textbook Chapter 11.4

Constructing public-key encryption

The 1976 Diffie-Hellman paper:

- constructed the key-exchange scheme we saw last time,
- defined public-key encryption,

BUT did not show how to build public key encryption!

They left that as an open problem.

With the benefit of hindsight……………….

Imagine that Bob wants to send a message M to Alice right away.

Public parameters: A big prime p, a big integer g

$\mathrm{sk}_A = a \in \{0, \ldots, p\text{-}1\}$

$K = g^{ab}$

$\mathrm{pk}_A = g^a \bmod p$

pk_B

Bob's favorite authenticated-encryption scheme
(E.g., AES-GCM)

$C = \mathrm{AE.Enc}(K, M)$

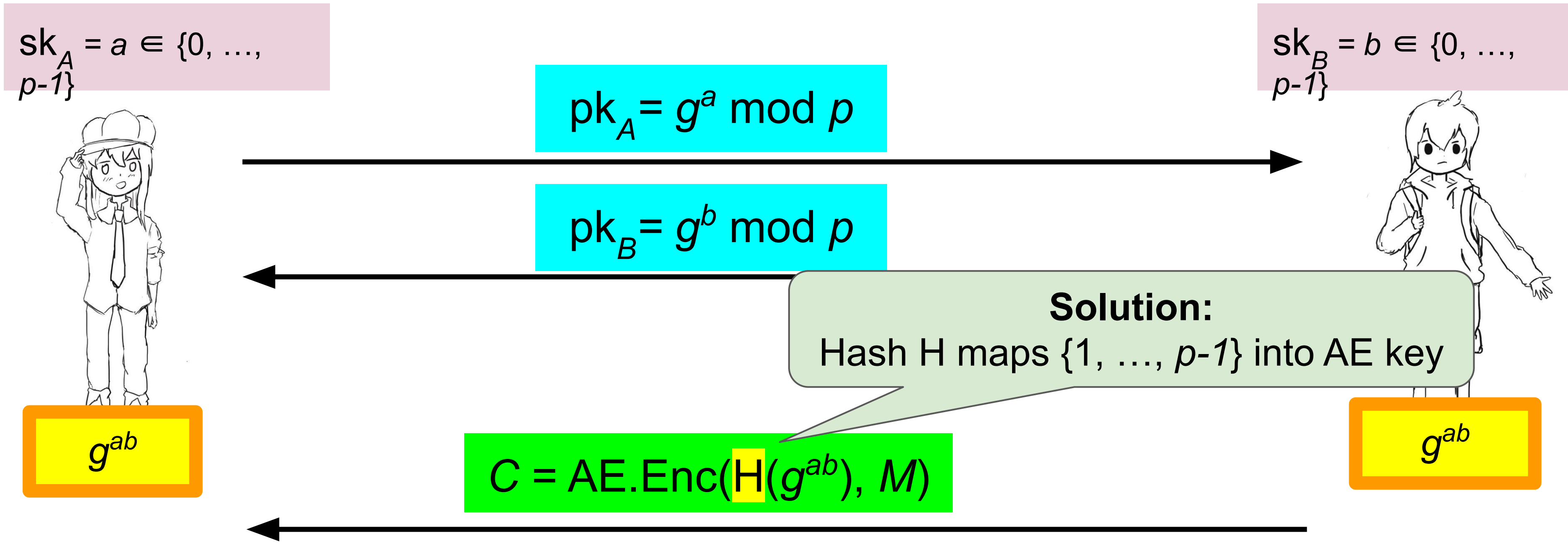
Problem: K is a number $\{1, \ldots, p\text{-}1\}$
We want an enc key, e.g., 128 bits

$\mathrm{sk}_B = b \in \{0, \ldots, p\text{-}1\}$

$K = g^{ab}$

Imagine that Bob wants to send a message M to Alice right away.

Public parameters: A big prime p, a big integer g



$\mathrm{sk}_A = a \in \{0, \ldots, p\text{-}1\}$

$\mathrm{sk}_B = b \in \{0, \ldots, p\text{-}1\}$

$\mathrm{pk}_A = g^a \bmod p$

$\mathrm{pk}_B = g^b \bmod p$

g^{ab}

g^{ab}

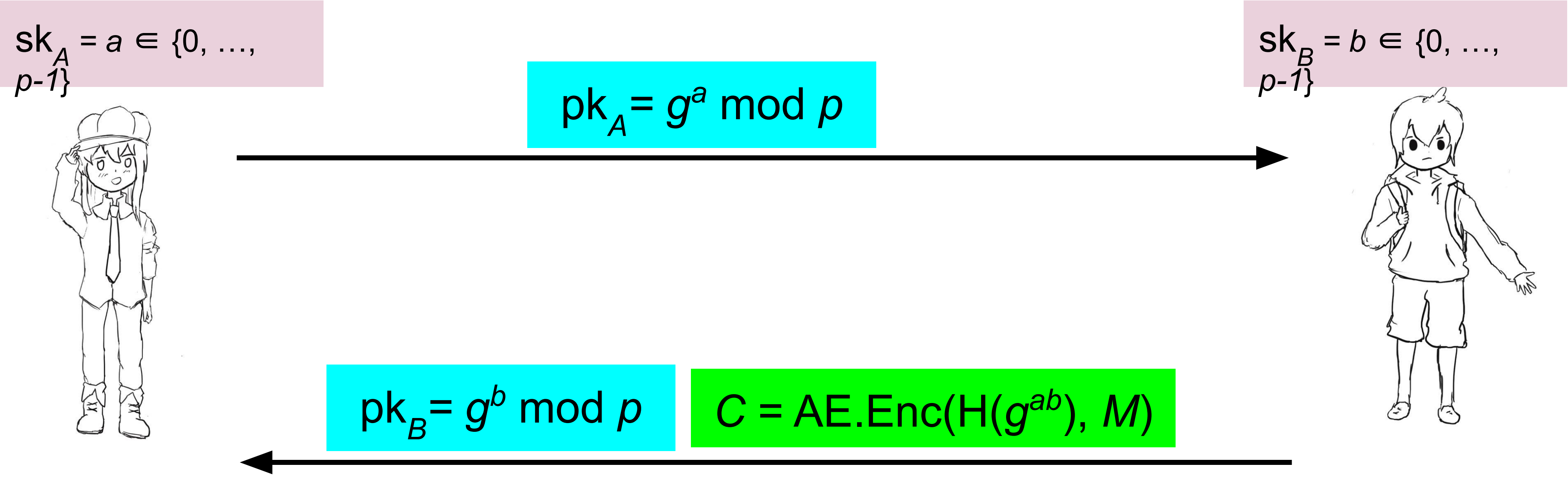
Solution:

Hash H maps $\{1, \ldots, p\text{-}1\}$ into AE key

$C = \mathrm{AE.Enc}(\mathrm{H}(g^{ab}), M)$

Diffie-Hellman key exchange

Public parameters: A big prime p, a big integer g



$\mathrm{sk}_A = a \in \{0, \ldots, p\text{-}1\}$

$\mathrm{sk}_B = b \in \{0, \ldots, p\text{-}1\}$

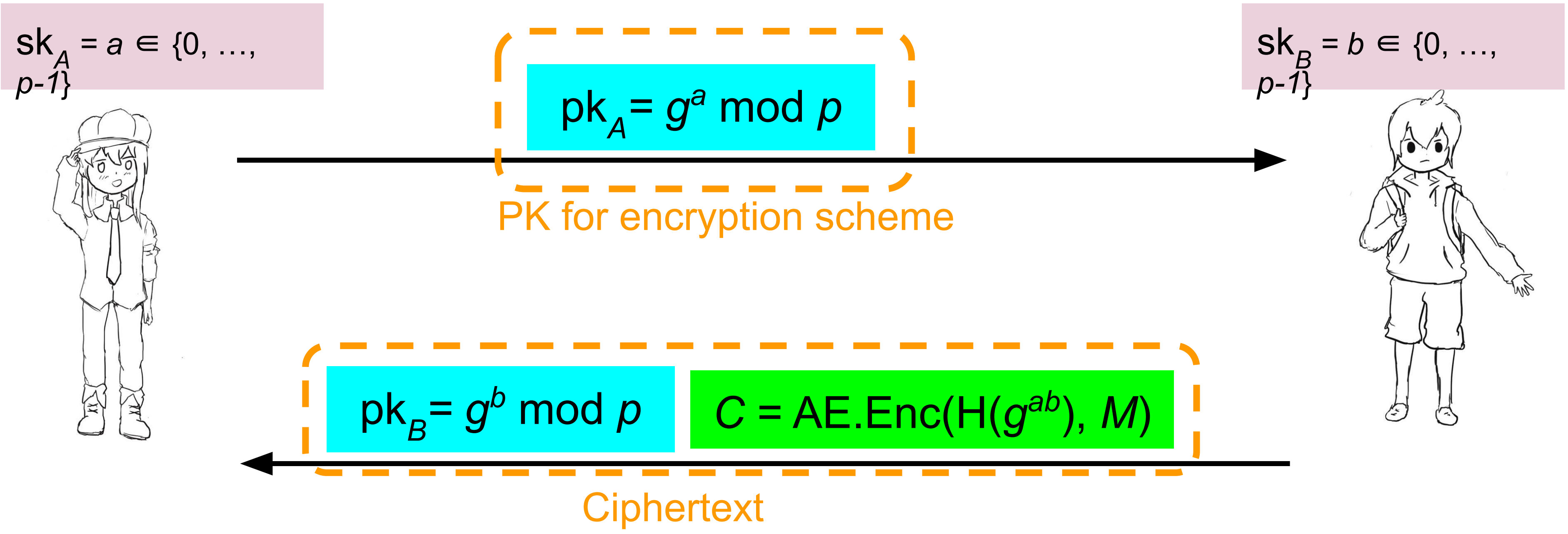
$\mathrm{pk}_A = g^a \bmod p$

$\mathrm{pk}_B = g^b \bmod p$

$C = \mathrm{AE.Enc}(\mathrm{H}(g^{ab}), M)$

Public-key encryption?

Public parameters: A big prime p, a big integer g



$\text{sk}_A = a \in \{0, \dots, p\text{-}1\}$

$\text{sk}_B = b \in \{0, \dots, p\text{-}1\}$

$\text{pk}_A = g^a \bmod p$

PK for encryption scheme

$\text{pk}_B = g^b \bmod p$

$C = \text{AE.Enc}(H(g^{ab}), M)$

Ciphertext

Hashed ElGamal Encryption

Parameters: Integers p, g used in DH key exchange
– Authenticated-encryption scheme (AE.Enc, AE.Dec) with keyspace $\{0,1\}^n$
– Hash function H : $\{1, \ldots, p-1\} \to \{0,1\}^n$

KeyGen() → (*SK*, *PK*): // Exactly as in DH key exchange

- Pick random a in $\{0, \ldots, p-1\}$
- Output key (a, g^a)

Encrypt($PK=g^a$, M):

- Sample random b in $\{0, \ldots, p-1\}$
- Compute $K \leftarrow \mathrm{H}(g^{ab})$
- Output $(g^b, \mathrm{AE.Enc}(K, M))$

Decrypt($SK=a$, $C=(C_0, C_1)$):

- // No randomness in decryption
- Compute $K \leftarrow \mathrm{H}(C_0{}^a)$ // $= \mathrm{H}(g^{ab})$
- Output $\mathrm{AE.Dec}(K, C_1)$

Security of ElGamal encryption

We will not prove:

Informal theorem. If:

- the computational Diffie-Hellman problem (CDH) is hard,
- the authenticated-encryption scheme (AE.Enc, AE.Dec) is CPA-secure, and
- we model the hash function H as an "ideal" hash function

then hashed ElGamal is CPA-secure.

ElGamal Encryption: Performance

Compute: 1 big-int exponentiation (expensive) + AE enc/dec (cheap)

On my laptop (Apple M3 Max):

DH: **6,700** (2048-bit $g^x \bmod p$) / second `$ openssl speed dsa`

ECDH: **348,230** (256-bit elliptic curve g^x) / second `$ openssl speed ecdh`

Auth enc/dec: **9.3 GB/s** `$ openssl speed -evp aes-128-gcm`

Communication: 1 DH public key + AE ciphertext

DH: = (3072 bits) + (message length) + (128-bit MAC tag) = $|M|$ + 400b

ECDH: = (256 bits) + (message length) + (128-bit MAC tag) = $|M|$ + 48b

Hybrid encryption

- This version of ElGamal is a “hybrid” encryption scheme:
  - One part is a **SLOW** public-key primitive (DH key exchange)
  - One part is a **FAST** secret-key primitive (authenticated encryption)
  - **SLOW** + **FAST** = It’s a hybrid of both

- There are non-hybrid public-key encryption schemes
  - If you want, you can have all **SLOW**
  - **SLOW** + **SLOW** + … + **SLOW** = Why would you want this?
  - Answer: More *functionality*

Classic ElGamal

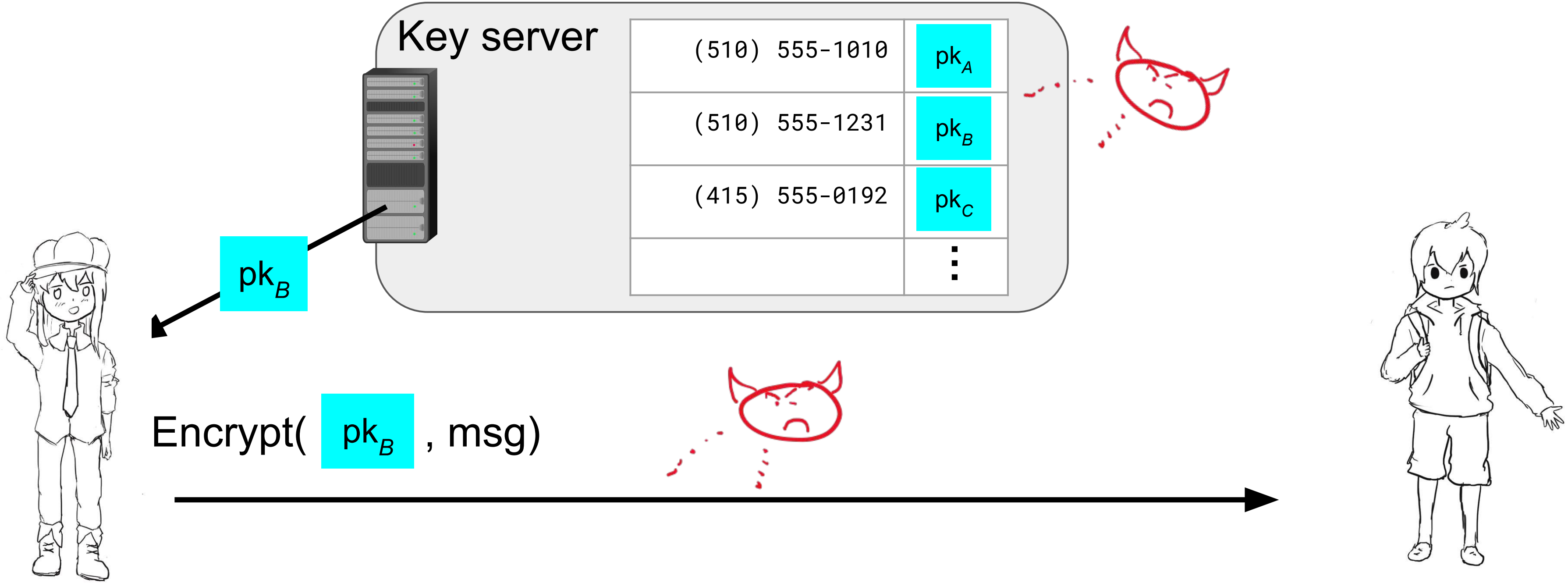
This is a variant of the ElGamal-encryption scheme we saw:
– Encrypts a small integer m in {0, …, 100}

$$\mathrm{Enc}(g^a, m) := (g^{ab}, g^{ab+m})$$ for random b in {0, …, p-1}

Cool feature: You can add messages “under encryption”

- $\mathrm{Enc}(g^a, m_0) + \mathrm{Enc}(g^a, m_1)$
  $= (g^{ab}, g^{ab+m0})+ (g^{ac}, g^{ac+m1})$
  $= (g^{a(b+c)}, g^{a(b+c)+m0+m1})$ // Let r = b+c
  $= (g^{ar}, g^{ar+m0+m1})$ = Encryption of $m_0 + m_1$
- Decryption requires solving discrete log ⇒ Only works for small messages
- “Additively homomorphic”
- Useful in some advanced applications

Public-key encryption



Key server

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(415) 555-0192	pk_C
	⋮

pk_B

Encrypt(pk_B , msg)

Problems we still need to solve

- **Integrity:** Adversary can tamper with messages from Alice to Bob
    - DH key exchange does not protect against this
    - Public-key encryption doesn't either—*not an authentication primitive*
    - Digital signatures = Very satisfying solution to this problem*
- Related issue: **What if key server is compromised?**
    - How does Alice know that she has Bob's public key (and not the attacker's)?
    - No totally satisfying solution exists (IMO)
    - Fancy cryptographic protections exist, Apple uses one ("transparency log")
- **Key server learns who is talking to whom!**
    - Apple, Signal, WhatsApp, etc., know who your contacts are
    - Fancy cryptographic solution exist, AFAIK none used in practice

RSA Encryption

- The first construction of public-key encryption
  - Very clever construction, requires non-trivial math
  - Like DH, still essentially unbroken after 50+ years
- At this point: inferior to elliptic-curve ElGamal in essentially all ways
  - Not used in any new protocols/schemes
- RSA signatures are still widely used
  - Also on the way out
  - But will be with us for a long time
- Hardness of factoring is necessary (not sufficient) for RSA to be secure
  - RSA gave a very strong practical reason to the study of factoring algorithms
- I will give you a taste of the RSA construction when we talk about signatures

Summary: Public-Key Encryption

- Public-key encryption: One key encrypts, the other decrypts
- ElGamal encryption
    - Combine DH key exchange with secret-key authenticated encryption
    - = Very fast, and also public key
    - Public-key encryption is rarely the performance bottleneck today
- Things we touched on:
    - Hybrid encryption
    - Additively homomorphic encryption = Adding encrypted messages
    - RSA Encryption

Next time: Digital signatures

Cryptography roadmap

	Secret-key crypto	**Public-key crypto**
Encryption	✅ One-time pad ✅ CPA security ✅ Counter-mode encryption ✅ CBC encryption ✅ Authenticated encryption	✅ Key exchange (e.g., Diffie-Hellman) ✅ Public-key encryption (e.g., ElGamal)
Authentication	✅ Message auth. codes (“MACs”)	– Digital signatures (e.g. RSA sigs)

Building blocks	✅ Pseudorandom functions ✅ Block ciphers (e.g., AES) ✅ Hash functions (e.g., SHA2)	✅ Discrete-log problem